

Final Presentation of Bachelor's Thesis

Design and Implementation of an 1 kVA Three-Phase GaN Inverter

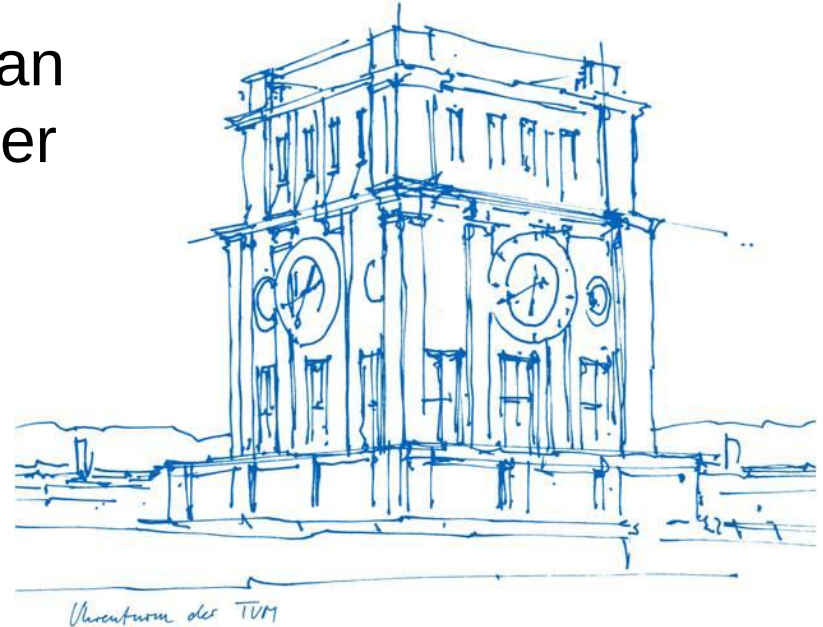
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23rd August, 2023

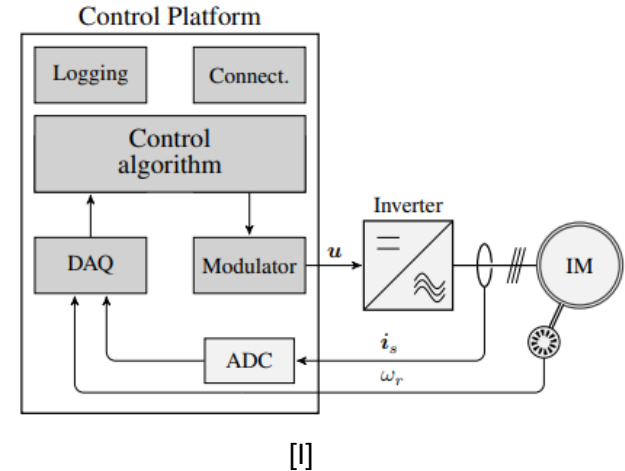


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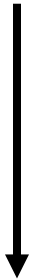
Motivation

- Using a three-phase two-level GaN inverter within UltraZohm
- Improving the gate driver circuit of the existing GaN inverter
- Optimizing the layout of the switching cells
- Implementing and commissioning
- Performance and thermal measurements



Gallium Nitride Power Transistor

MOSFET Inverter



GaN Inverter

Benefits:

- Higher switching speed
- Higher efficiency
- Lower $R_{DS(ON)}$
- Thermally enhanced package
- Smaller footprint, high power density

Parameters	EPC2302
V_{DS}	100 V
I_D	101 A
$R_{DS(on)} \text{ max.}$	1.8 m Ω
V_{GS}	−4 V to 6 V
C_{iss}	3200 pF
Q_G	23-29 nC
Q_{RR}	0 nC
$R_{\theta JC}$	0.2 $\frac{^{\circ}\text{C}}{\text{W}}$
Size	15 mm ²

Three-phase Two-level GaN Inverter

- Absolute maximum ratings:

$$V_{dc} = 48 \text{ V} \quad I_{out} = 25 \text{ A}$$

$$f_{PWM} = 100 \text{ kHz} \quad T_{max} = 150^{\circ}\text{C}$$

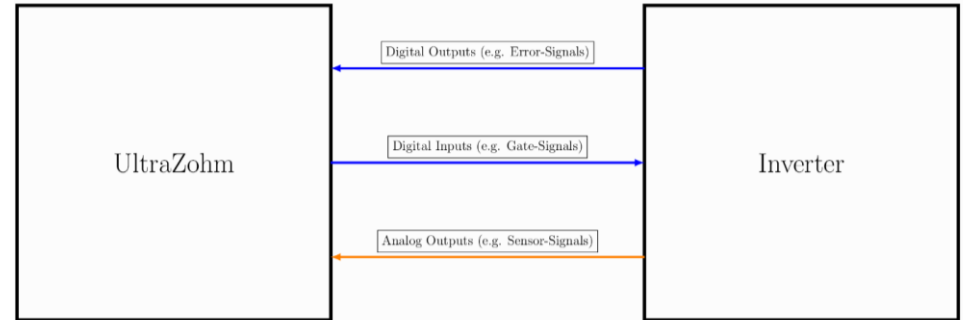
- Current and voltage measurements:

$$I_{OCP, leg} = 40 \text{ A} \quad I_{OCP, phase} = 29.85 \text{ A}$$

$$V_{peak} = 60 \text{ V}$$

- Temperature measurement of GaN HEMTs

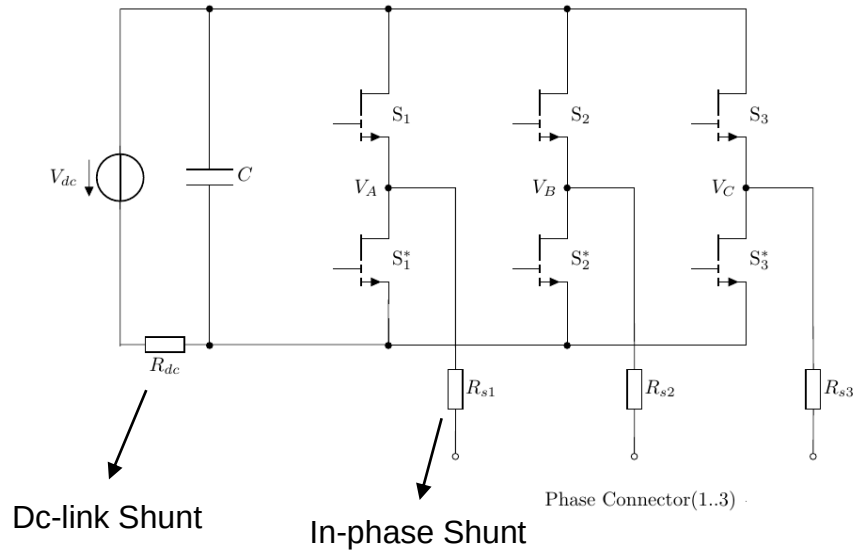
$$T_{peak} = 105^{\circ}\text{C}$$



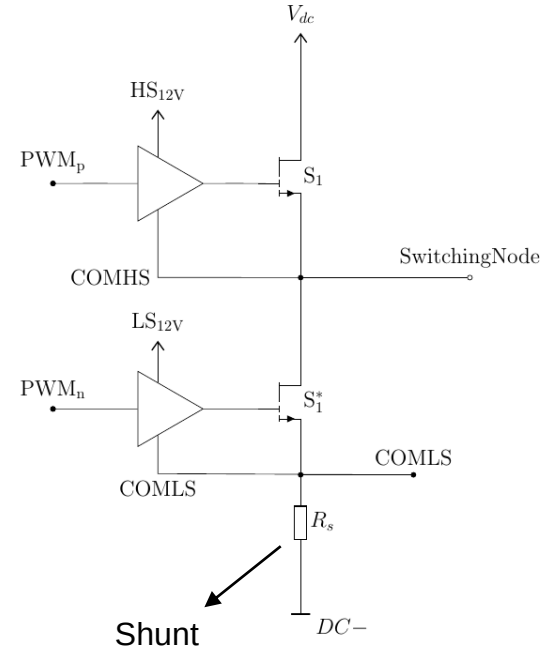
[11]

Three-phase Two-level GaN Inverter

Three Phase Inverter without Gate Drivers

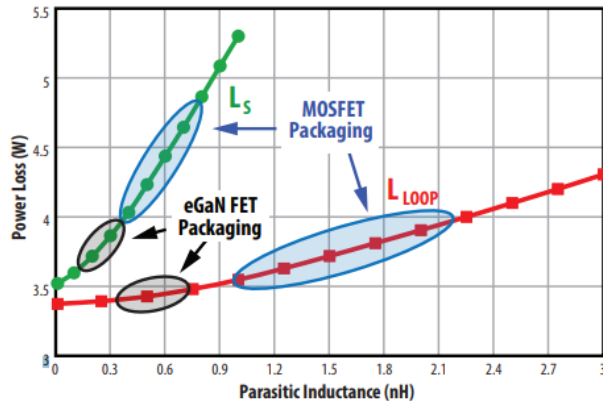


Single-leg with Gate Drivers & Shunt Resistors



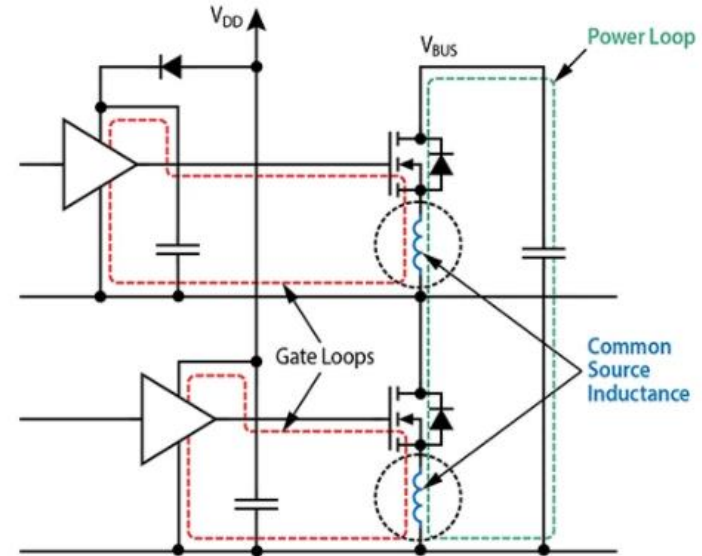
Parasitic Inductances

- 1) Common-Source Inductance (L_s)
- 2) Power Loop Inductance (L_{loop})
- 3) Gate Loop Inductance (L_g)



Parasitic inductance impact on power loss ($V_{IN} = 12\text{ V}$, $V_{OUT} = 1.2\text{ V}$, $I_{OUT} = 20\text{ A}$, $F_{SW} = 1\text{ MHz}$)

[IX]



Schematic of a half-bridge power stage showing power and gate drive loops. [IV]

Optimal GaN FET Layout

- L_s is mainly controlled by the package inductance
- Overall reduction in parasitic inductance through magnetic field self-cancellation
 CSI is minimized
- L_s is no longer major loss contributor, but L_{loop}

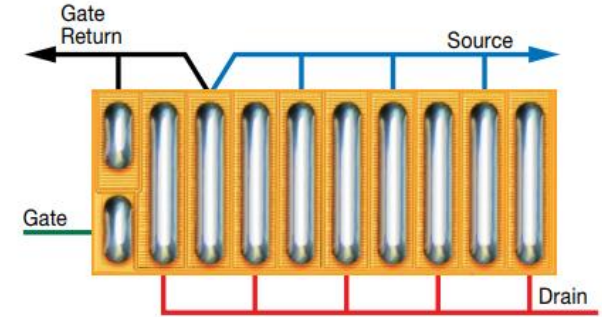
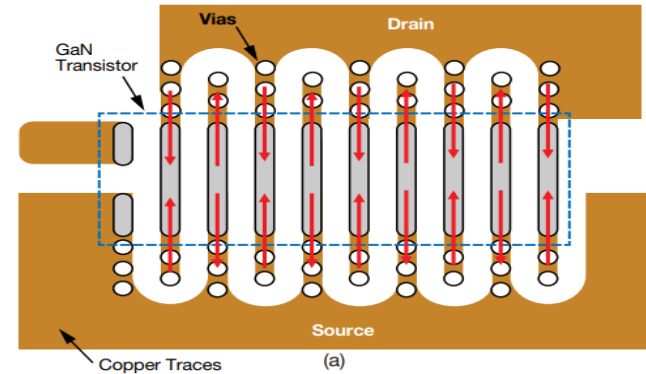


Figure 1: GaN transistor in an LGA format showing the direction of device current flow that minimizes common source inductance.

[VII]

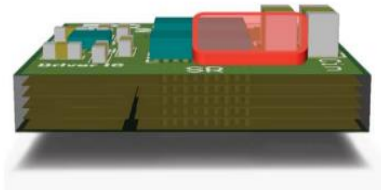
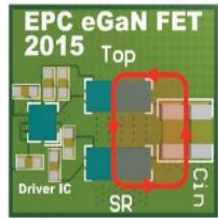


(a)

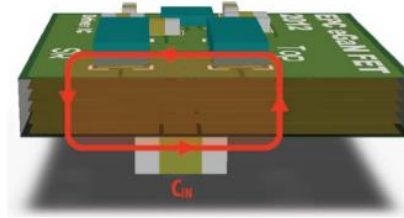
[VII]

Comparison of Power Loop Layouts

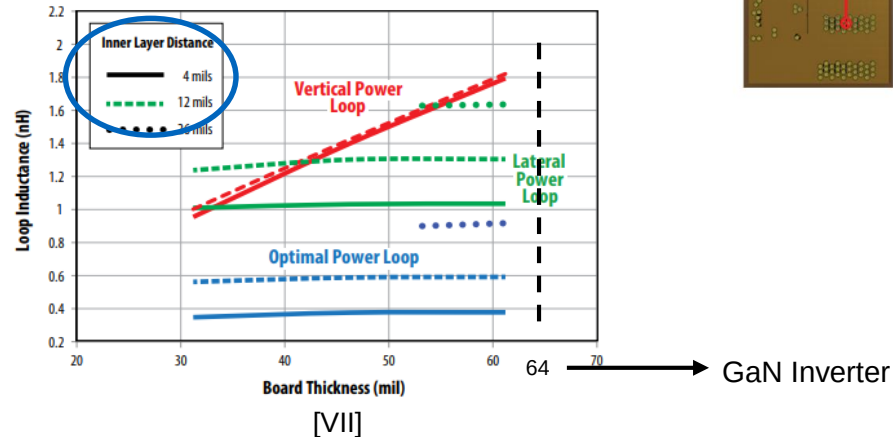
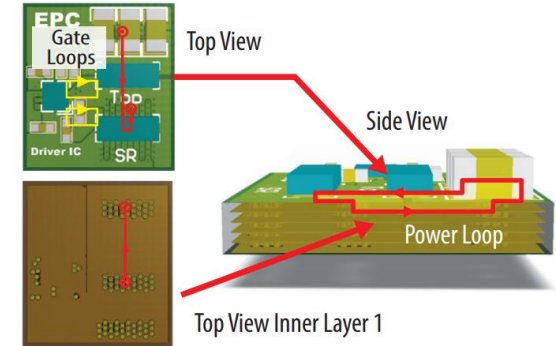
1) Conventional Lateral Power Loop



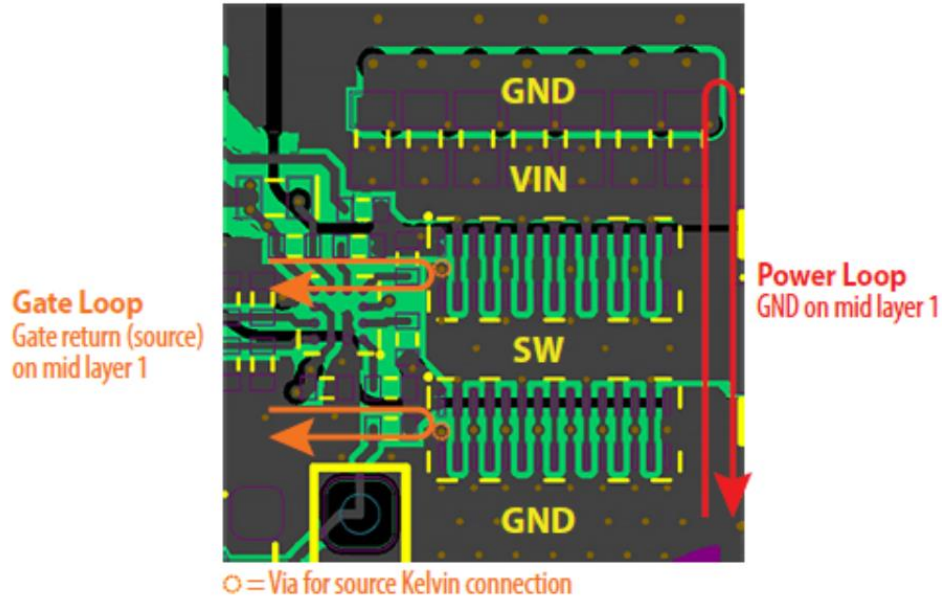
2) Conventional Vertical Power Loop



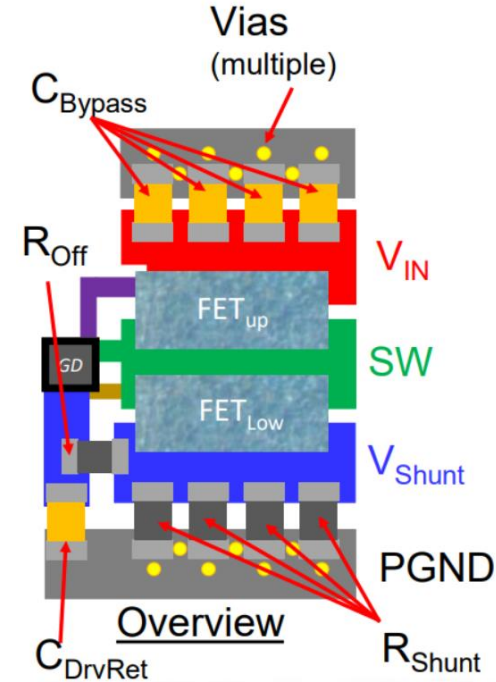
3) Proposed Optimal Power Loop



Optimal Layout of Power and Gate Loops



[V]



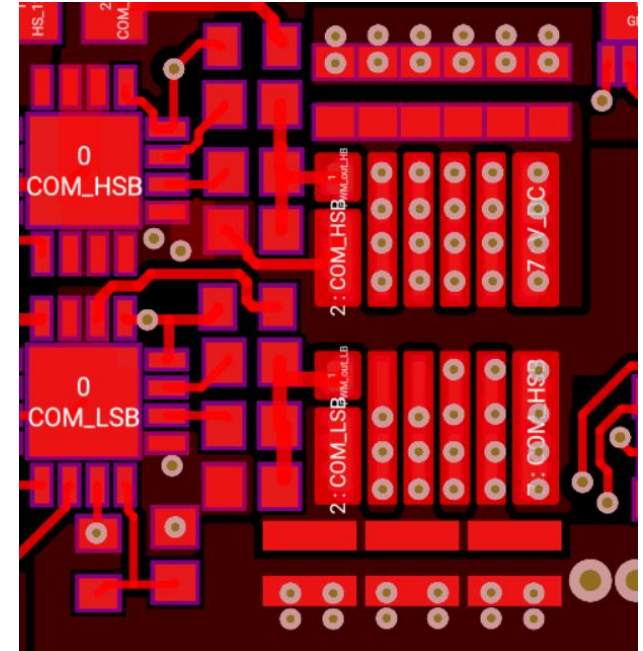
[III]

Vias Plated Over

- Vias must be filled with epoxy and high thermal conductivity copper
- Minimizes the signal path lengths, decreasing the parasitics
- Costly and complex to produce

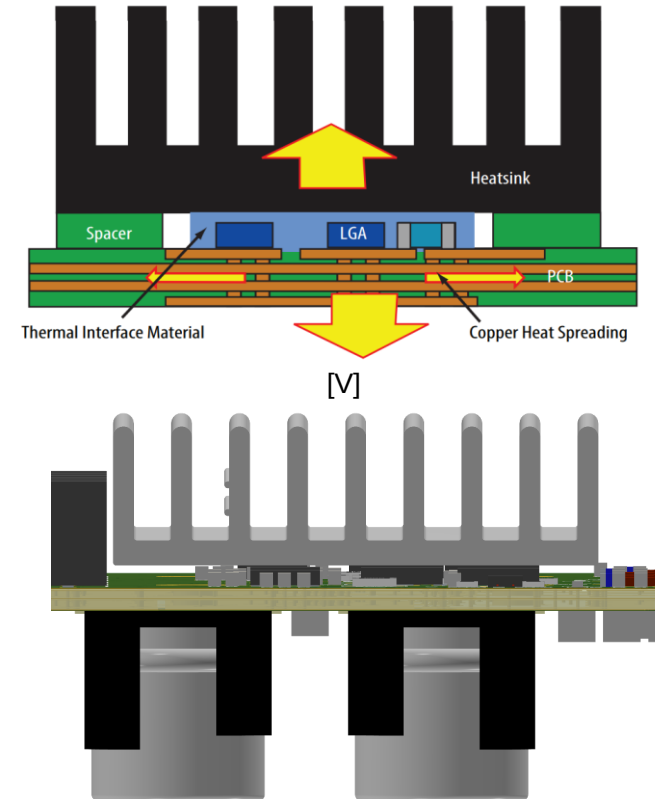


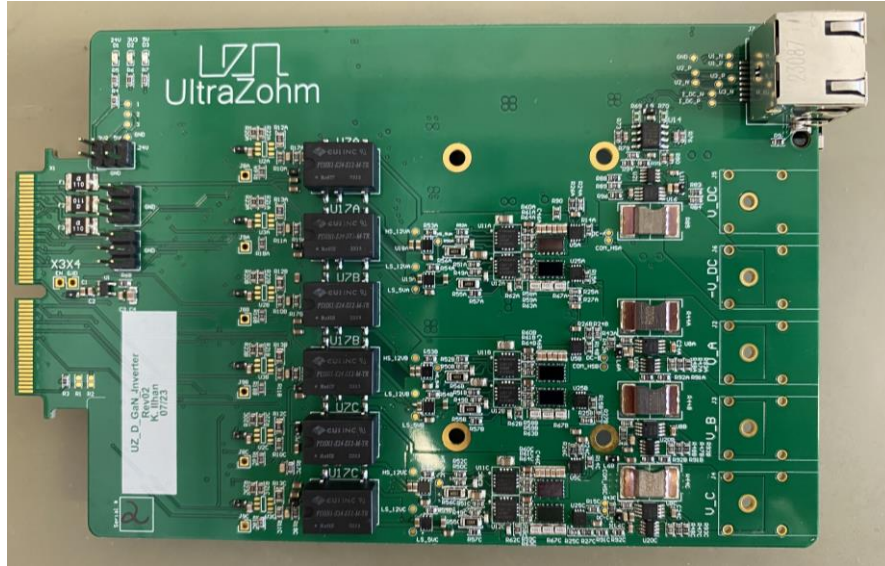
[VIII]



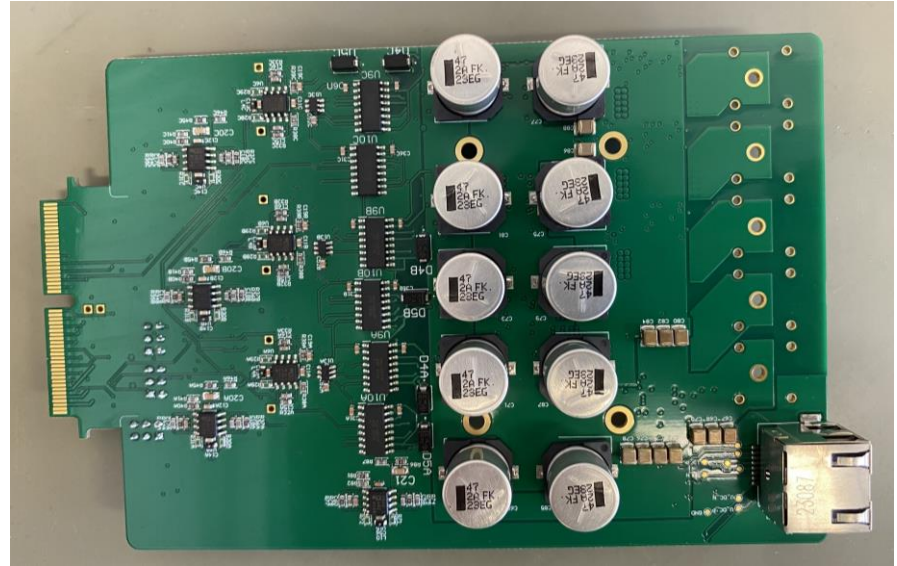
Thermal Considerations

- Prepared for a heatsink installation
- Four holes of M3 are placed to allow screws
- Careful placement of thermal vias and power planes to spread the heat
- Epoxy-filled vias to improve heat dissipation





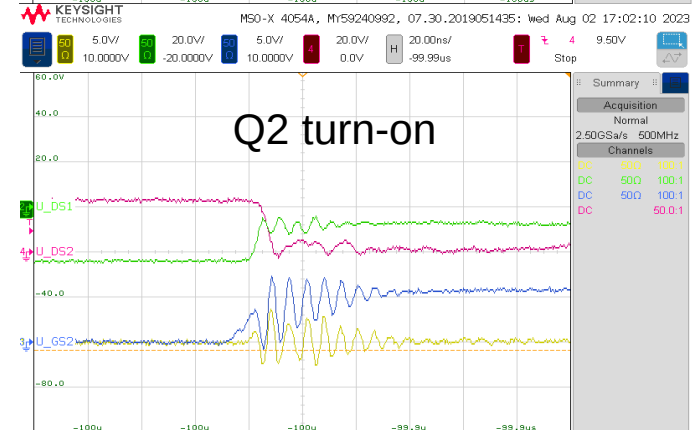
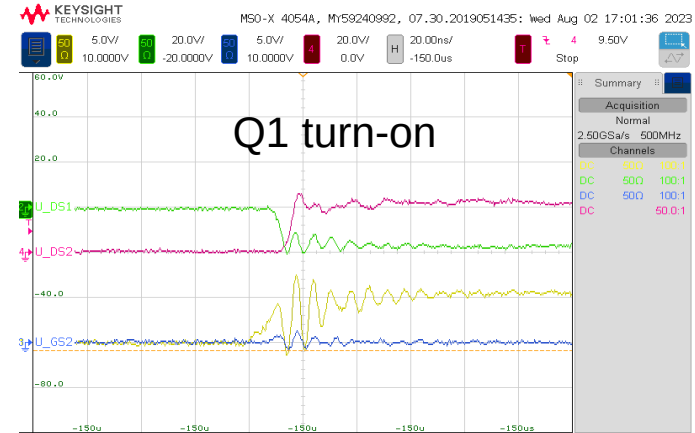
Top view



Bottom view

Measurement Results (M3)

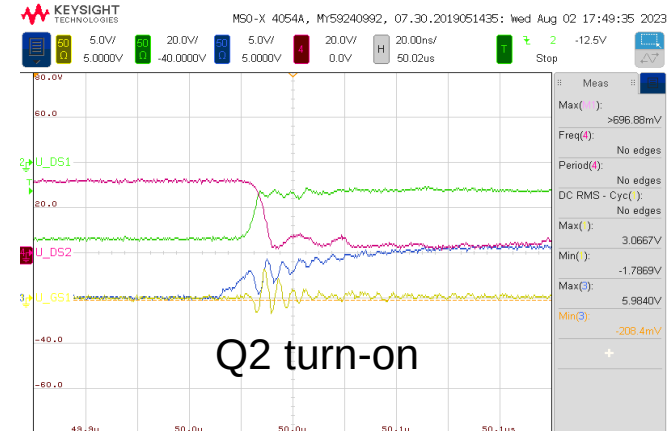
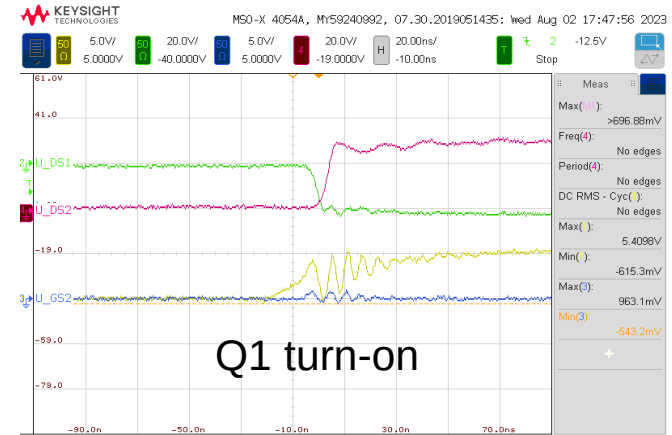
- $V_{GS1} \rightarrow$ yellow
 - $V_{DS1} \rightarrow$ green
 } Q1 (top switch)
- $V_{GS2} \rightarrow$ blue
 - $V_{DS2} \rightarrow$ pink
 } Q2 (bottom switch)
- $V_{dc} = 24\text{ V}, f_{PWM} = 10\text{ kHz}$
- $I_{SRC} = 2\text{ A}, R_{ON} = 2.7\ \Omega, R_{OFF} = 2.7\ \Omega$



Measurement Results (M4)

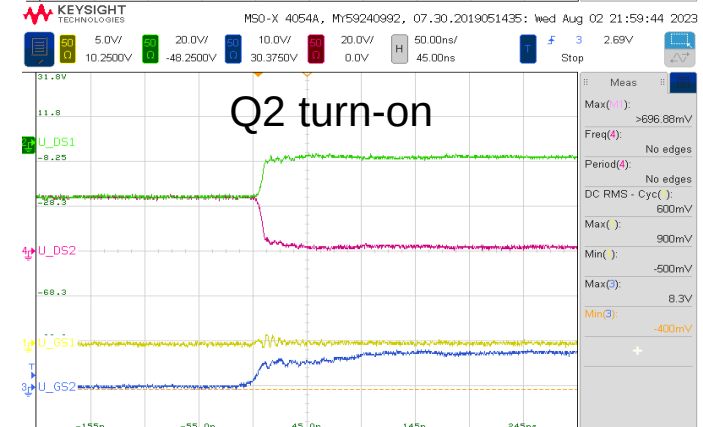
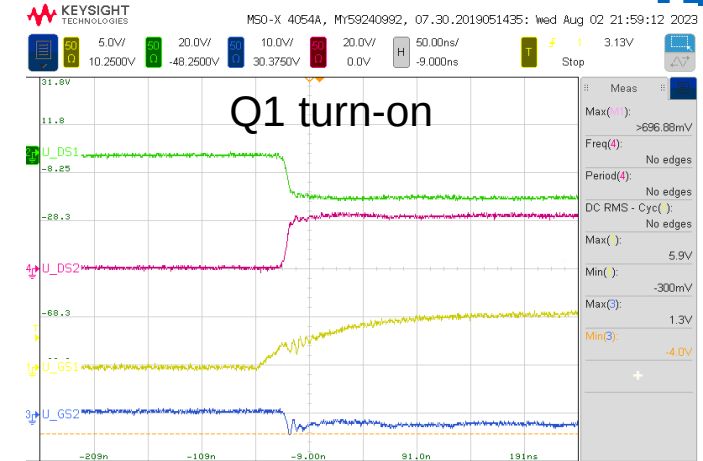
- $V_{GS1} \rightarrow$ yellow
 - $V_{DS1} \rightarrow$ green
 } Q1 (top switch)
- $V_{GS2} \rightarrow$ blue
 - $V_{DS2} \rightarrow$ pink
 } Q2 (bottom switch)

- $V_{dc} = 34\text{ V}$, $f_{PWM} = 10\text{ kHz}$
- $I_{SRC} = 0.75\text{ A}$, $R_{ON} = 2.7\text{ }\Omega$, $R_{OFF} = 2.7\text{ }\Omega$,



Measurement Results (M5)

- $V_{GS1} \rightarrow$ yellow
 - $V_{DS1} \rightarrow$ green
 } Q1 (top switch)
- $V_{GS2} \rightarrow$ blue
 - $V_{DS2} \rightarrow$ pink
 } Q2 (bottom switch)
- $V_{dc} = 24\text{ V}$, $f_{PWM} = 10\text{ kHz}$
- $I_{SRC} = 0.75\text{ A}$, $R_{ON} = 10\text{ }\Omega$, $R_{OFF} = 2.7\text{ }\Omega$,



Thermal Measurements

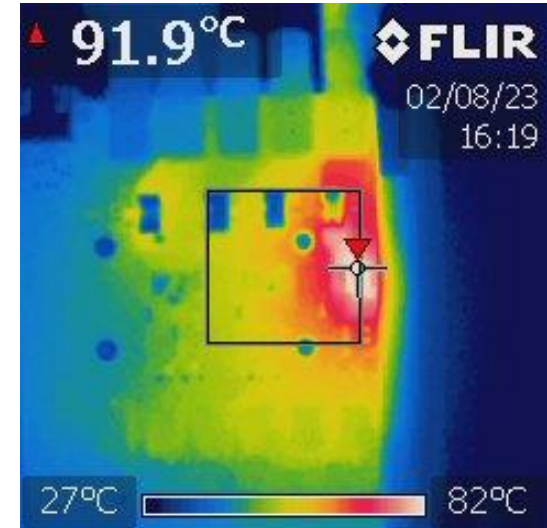
	8-10 A	15 A	19 A	21 A	24-25 A
24 V; 10 kHz	48.4 °C	56.8 °C	74.9 °C	78.8 °C	81.2 °C
48 V; 10 kHz	51.7 °C	59.9 °C	72.7 °C	N/A	91 °C
24 V; 50 kHz	72 °C	96 °C	N/A	N/A	N/A
24 V; 100 kHz	93.4 °C	N/A	N/A	N/A	N/A

Temperatures of the GaN devices at various measurement points **without heat sink**

with $I_{SRC} = 0.75 \text{ A}$, $R_{ON} = 10 \text{ } \Omega$, $R_{OFF} = 2.7 \text{ } \Omega$, dead-time = $1 \text{ } \mu\text{s}$

Reasons:

- Slow switching speed with the selected values of I_{SRC} and R_{ON}
 Finding an optimal compromise
- Dead-time value



Temperature measurement with $I_{OUT} = 24 \text{ A}$ at $f_{PWM} = 10 \text{ kHz}$, $V_{dc} = 48 \text{ V}$, $I_{SRC} = 0.75 \text{ A}$ and $R_{ON} = 10 \text{ } \Omega$, $R_{OFF} = 2.7 \text{ } \Omega$, dead-time = $1 \text{ } \mu\text{s}$

References:

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